

DATA FLOW DIAGRAM

EDUGENIE – AI-POWERED LEARNING ASSISTANT

DATA FLOW DIAGRAM – TEXTUAL REPRESENTATION

LEVEL 0 – OVERALL FLOW

User



EduGenie Web Interface



FastAPI Backend



Selected Learning Module



AI Model / Service



Generated Result



FastAPI Backend



EduGenie Web Interface



User

MODULE-SPECIFIC FLOW

Question Answering:

User Question → FastAPI → Q&A Module → Gemini 3.5 Flash → Answer → Frontend

Concept Explanation:

User Topic → FastAPI → Explanation Module → LaMini-Flan-T5-783M → Simplified Explanation → Frontend

Quiz:

Topic / Material → FastAPI → Quiz Module → Gemini 3.5 Flash → Multiple-Choice Quiz → Frontend

Summary:

Long Text → FastAPI → Summary Module → Gemini 3.5 Flash → Concise Summary → Frontend

Learning Path:

Learning Topic → FastAPI → Learning Path Module → Gemini 3.5 Flash → Structured Learning Path → Frontend

CONFIGURATION FLOW

Gemini API Credential → Environment Variable → Gemini Client → Gemini Service